Gravitational field strength

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a gravitational field to predict the force on a test mass.

SUCCESS CRITERIA - I can...

1. Represent field direction with arrows.
2. Use g=F/m and F=mg consistently.
3. Distinguish source field from the mass placed in it.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “field” and “gravity” to explain your model or result. Sentence starters are optional.

field: a description of an influence at each location

gravity: an attractive interaction between masses

radius: distance from the center to the edge of a circle or sphere

01 Bring back an earlier idea

A 3 kg object is in a field of 5 N/kg.
Give its mass and weight, and distinguish the two quantities.

Word help: weight = the gravitational force on an object

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $g = F_g/m = GM/r^2$.
Outside a spherical source of mass M; r is distance from its center. Field direction is toward the source.

Worked reasoning: A 2 kg test object feels 6 N toward a source. The local field is g=6/2=3 N/kg. A 4 kg test object at that point feels 12 N.

Your turn: Draw the field arrow and the two possible force arrows. Which quantity belongs to the location rather than the test mass?

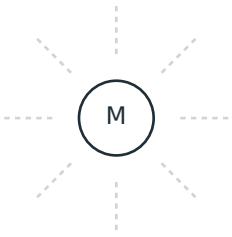
Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

03 Look and explain

Draw field arrows at three locations around the source. Which way would a small test mass accelerate?

Sentence starter: The … in my model represents … . Its label shows … .



Add arrowheads to show field direction.

The center marks the source; add arrowheads on the surrounding lines to show field direction.

Marzano check - circle: ① ② ③

04 Build the solution

A field is 4 N/kg.
Find the gravitational force on 1 kg, 2 kg and 3 kg objects. Does changing test mass change the given field?

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

A student says a heavier test mass falls faster because it feels more gravitational force.
Use F=mg and a=F/m to respond.

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: 1 2 3

06 Choose a method

A 4 kg cart moves left at 6 m/s. Forces are 38 N right and 6 N left.
Find acceleration and whether its speed initially rises or falls.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

07 Exit check

A 2 kg test mass feels 10 N.
Find g and the force on 4 kg there.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Two test masses experience forces in a 1: 3 ratio at the same point. Create possible masses and force values for a 5 N/kg field.

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: 1 2 3